

Contracted Bethe–Heitler Helicity Amplitudes in User-Frame Kinematics

$$\theta_{\text{in}} = \pi/2$$

Helicity labels are doubled helicities in $\{-1, +1\}$. The electron is massless, so $h' = h$ and every amplitude with $h' \neq h$ vanishes.

User-frame kinematics

Define $\alpha = \phi_{\text{in}}$, $\varphi = \phi_{\text{Out}}$, $w = q_{\text{Out}}$, and

$$\begin{aligned} P &= \frac{s - m^2}{2\sqrt{s}}, & E &= \sqrt{P^2 + m^2}, \\ R &= p_{\text{Out}}, & E' &= \sqrt{R^2 + m^2}, \\ K &= \sqrt{R^2 + w^2 + 2Rw \sin \varphi}, & \sqrt{E'^2 - m^2} &= R. \end{aligned}$$

Energy conservation fixes R through $E' + K + w = \sqrt{s}$. The external momenta are

$$\begin{aligned} k^\mu &= P(1, -\cos \alpha, -\sin \alpha, 0), \\ p^\mu &= (E, P \cos \alpha, P \sin \alpha, 0), \\ k'^\mu &= (K, -w \cos \varphi, -R - w \sin \varphi, 0), \\ p'^\mu &= (E', 0, R, 0), \\ q'^\mu &= w(1, \cos \varphi, \sin \varphi, 0). \end{aligned}$$

The momentum products that contain every propagator denominator are

$$\begin{aligned} D_+ &\equiv 2k' \cdot q' = 2w(K + w + R \sin \varphi), \\ D_- &\equiv -2k \cdot q' = -2Pw[1 + \cos(\alpha - \varphi)], \\ t &= (p' - p)^2 = 2m^2 - 2EE' + 2PR \sin \alpha. \end{aligned}$$

To shorten phases without hiding kinematics, define only

$$z_a = e^{i\alpha}, \quad \bar{z}_a = e^{-i\alpha}, \quad z_f = e^{i\varphi}, \quad \bar{z}_f = e^{-i\varphi}.$$

Fully contracted scalar building blocks

After all Dirac and polarization contractions, write

$$\mathcal{L}_{h\lambda}^\nu = \frac{N_{h\lambda}^{\nu,+}}{D_+} + \frac{N_{h\lambda}^{\nu,-}}{D_-}, \quad \mathcal{H}_\nu^{s's} = \bar{u}(p', s') \Gamma_\nu u(p, s).$$

The following equations are explicit scalars; no gamma matrices or spinors remain.

Leptonic numerators

$$h = -1, \lambda = -1$$

$$\begin{aligned}
N_{-1,-1}^{0,+} &= \sqrt{2}\sqrt{P} \left(\frac{i\sqrt{K}Rz_f}{2} + \frac{i\sqrt{K}R\bar{z}_f}{2} - \sqrt{K}w\bar{z}_az_f + \sqrt{K}w - \frac{R^2\bar{z}_az_f}{2\sqrt{K}} - \frac{R^2\bar{z}_a\bar{z}_f}{2\sqrt{K}} + \frac{iRw\bar{z}_az_f^2}{2\sqrt{K}} - \frac{iRw\bar{z}_a}{2\sqrt{K}} + \frac{iRw\bar{z}_f}{\sqrt{K}} - \frac{w^2\bar{z}_az_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right) \\
N_{-1,-1}^{0,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(\sqrt{K}z_a\bar{z}_f - \sqrt{K}\bar{z}_az_f - \frac{iR\bar{z}_a^2z_f}{\sqrt{K}} + \frac{iR\bar{z}_f}{\sqrt{K}} - \frac{w\bar{z}_a^2z_f^2}{\sqrt{K}} + \frac{w}{\sqrt{K}} \right)}{2} \\
N_{-1,-1}^{1,+} &= \sqrt{2}\sqrt{P} \left(-\frac{i\sqrt{K}R\bar{z}_az_f}{2} - \frac{i\sqrt{K}R\bar{z}_a\bar{z}_f}{2} - \sqrt{K}w\bar{z}_a + \sqrt{K}wz_f + \frac{R^2z_f}{2\sqrt{K}} + \frac{R^2\bar{z}_f}{2\sqrt{K}} - \frac{iRw\bar{z}_a\bar{z}_f}{\sqrt{K}} - \frac{iRwz_f^2}{2\sqrt{K}} + \frac{iRw}{2\sqrt{K}} - \frac{w^2\bar{z}_a}{\sqrt{K}} + \frac{w^2z_f}{\sqrt{K}} \right) \\
N_{-1,-1}^{1,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(\sqrt{K}\bar{z}_a^2z_f - \sqrt{K}\bar{z}_f - \frac{iRz_a\bar{z}_f}{\sqrt{K}} + \frac{iR\bar{z}_az_f}{\sqrt{K}} - \frac{wz_a}{\sqrt{K}} + \frac{w\bar{z}_az_f^2}{\sqrt{K}} \right)}{2} \\
N_{-1,-1}^{2,+} &= \sqrt{2}\sqrt{P} \left(\frac{\sqrt{K}R\bar{z}_az_f}{2} + \frac{\sqrt{K}R\bar{z}_a\bar{z}_f}{2} - i\sqrt{K}w\bar{z}_a - i\sqrt{K}wz_f - \frac{iR^2z_f}{2\sqrt{K}} - \frac{iR^2\bar{z}_f}{2\sqrt{K}} + \frac{Rw\bar{z}_a\bar{z}_f}{\sqrt{K}} - \frac{Rwz_f^2}{2\sqrt{K}} + \frac{Rw}{2\sqrt{K}} - \frac{iw^2\bar{z}_a}{\sqrt{K}} - \frac{iw^2z_f}{\sqrt{K}} \right) \\
N_{-1,-1}^{2,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(i\sqrt{K}\bar{z}_a^2z_f - i\sqrt{K}\bar{z}_f - \frac{Rz_a\bar{z}_f}{\sqrt{K}} + \frac{R\bar{z}_az_f}{\sqrt{K}} + \frac{iwz_a}{\sqrt{K}} - \frac{iw\bar{z}_az_f^2}{\sqrt{K}} \right)}{2} \\
N_{-1,-1}^{3,+} &= \sqrt{2}\sqrt{P} \left(\frac{i\sqrt{K}Rz_f}{2} + \frac{i\sqrt{K}R\bar{z}_f}{2} + \sqrt{K}w\bar{z}_az_f + \sqrt{K}w + \frac{R^2\bar{z}_az_f}{2\sqrt{K}} + \frac{R^2\bar{z}_a\bar{z}_f}{2\sqrt{K}} - \frac{iRw\bar{z}_az_f^2}{2\sqrt{K}} + \frac{iRw\bar{z}_a}{2\sqrt{K}} + \frac{iRw\bar{z}_f}{\sqrt{K}} + \frac{w^2\bar{z}_az_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right) \\
N_{-1,-1}^{3,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(\sqrt{K}z_a\bar{z}_f - \sqrt{K}\bar{z}_az_f + \frac{iR\bar{z}_a^2z_f}{\sqrt{K}} - \frac{iR\bar{z}_f}{\sqrt{K}} + \frac{w\bar{z}_a^2z_f^2}{\sqrt{K}} - \frac{w}{\sqrt{K}} \right)}{2}
\end{aligned}$$

$$h = -1, \lambda = 1$$

$$N_{-1,1}^{0,+} = \frac{\sqrt{2}\sqrt{P}R \left(-i\sqrt{K}z_f - i\sqrt{K}\bar{z}_f + \frac{R\bar{z}_a z_f}{\sqrt{K}} + \frac{R\bar{z}_a \bar{z}_f}{\sqrt{K}} - \frac{iw\bar{z}_a z_f^2}{\sqrt{K}} - \frac{iw\bar{z}_a}{\sqrt{K}} \right)}{2}$$

$$N_{-1,1}^{0,-} = \sqrt{2}\sqrt{P} \left(-\frac{\sqrt{K}Pz_a \bar{z}_f}{2} + \frac{\sqrt{K}P\bar{z}_a z_f}{2} - \sqrt{K}w\bar{z}_a z_f - \sqrt{K}w + \frac{iPR\bar{z}_a^2 z_f}{2\sqrt{K}} - \frac{iPR\bar{z}_f}{2\sqrt{K}} + \frac{Pw\bar{z}_a^2 z_f^2}{2\sqrt{K}} - \frac{Pw}{2\sqrt{K}} + \frac{iRw\bar{z}_a}{\sqrt{K}} + \frac{iRw\bar{z}_f}{\sqrt{K}} + \frac{w^2\bar{z}_a z_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right)$$

$$N_{-1,1}^{1,+} = \frac{\sqrt{2}\sqrt{P}R \left(i\sqrt{K}\bar{z}_a z_f + i\sqrt{K}\bar{z}_a \bar{z}_f - \frac{Rz_f}{\sqrt{K}} - \frac{R\bar{z}_f}{\sqrt{K}} + \frac{iwz_f^2}{\sqrt{K}} + \frac{iw}{\sqrt{K}} \right)}{2}$$

$$N_{-1,1}^{1,-} = \sqrt{2}\sqrt{P} \left(-\frac{\sqrt{K}P\bar{z}_a^2 z_f}{2} + \frac{\sqrt{K}P\bar{z}_f}{2} - \sqrt{K}w\bar{z}_a - \sqrt{K}w\bar{z}_f + \frac{iPRz_a \bar{z}_f}{2\sqrt{K}} - \frac{iPR\bar{z}_a z_f}{2\sqrt{K}} + \frac{Pwz_a}{2\sqrt{K}} - \frac{Pw\bar{z}_a z_f^2}{2\sqrt{K}} + \frac{iRw\bar{z}_a z_f}{\sqrt{K}} + \frac{iRw}{\sqrt{K}} + \frac{w^2\bar{z}_a z_f^2}{\sqrt{K}} + \frac{w^2 z_f}{\sqrt{K}} \right)$$

$$N_{-1,1}^{2,+} = \frac{\sqrt{2}\sqrt{P}R \left(-\sqrt{K}\bar{z}_a z_f - \sqrt{K}\bar{z}_a \bar{z}_f + \frac{iRz_f}{\sqrt{K}} + \frac{iR\bar{z}_f}{\sqrt{K}} + \frac{wz_f^2}{\sqrt{K}} + \frac{w}{\sqrt{K}} \right)}{2}$$

$$N_{-1,1}^{2,-} = \sqrt{2}\sqrt{P} \left(-\frac{i\sqrt{K}P\bar{z}_a^2 z_f}{2} + \frac{i\sqrt{K}P\bar{z}_f}{2} - i\sqrt{K}w\bar{z}_a - i\sqrt{K}w\bar{z}_f + \frac{PRz_a \bar{z}_f}{2\sqrt{K}} - \frac{PR\bar{z}_a z_f}{2\sqrt{K}} - \frac{iPwz_a}{2\sqrt{K}} + \frac{iPw\bar{z}_a z_f^2}{2\sqrt{K}} + \frac{Rw\bar{z}_a z_f}{\sqrt{K}} + \frac{Rw}{\sqrt{K}} - \frac{iw^2\bar{z}_a z_f^2}{\sqrt{K}} - \frac{iw^2 z_f}{\sqrt{K}} \right)$$

$$N_{-1,1}^{3,+} = \frac{\sqrt{2}\sqrt{P}R \left(-i\sqrt{K}z_f - i\sqrt{K}\bar{z}_f - \frac{R\bar{z}_a z_f}{\sqrt{K}} - \frac{R\bar{z}_a \bar{z}_f}{\sqrt{K}} + \frac{iw\bar{z}_a z_f^2}{\sqrt{K}} + \frac{iw\bar{z}_a}{\sqrt{K}} \right)}{2}$$

$$N_{-1,1}^{3,-} = \sqrt{2}\sqrt{P} \left(-\frac{\sqrt{K}Pz_a \bar{z}_f}{2} + \frac{\sqrt{K}P\bar{z}_a z_f}{2} - \sqrt{K}w\bar{z}_a z_f - \sqrt{K}w - \frac{iPR\bar{z}_a^2 z_f}{2\sqrt{K}} + \frac{iPR\bar{z}_f}{2\sqrt{K}} - \frac{Pw\bar{z}_a^2 z_f^2}{2\sqrt{K}} + \frac{Pw}{2\sqrt{K}} - \frac{iRw\bar{z}_a}{\sqrt{K}} - \frac{iRw\bar{z}_f}{\sqrt{K}} - \frac{w^2\bar{z}_a z_f}{\sqrt{K}} - \frac{w^2}{\sqrt{K}} \right)$$

$$h = 1, \lambda = -1$$

$$N_{1,-1}^{0,+} = \frac{\sqrt{2}\sqrt{P}R \left(i\sqrt{K}z_f + i\sqrt{K}\bar{z}_f + \frac{Rz_az_f}{\sqrt{K}} + \frac{Rz_a\bar{z}_f}{\sqrt{K}} + \frac{iwz_a\bar{z}_f^2}{\sqrt{K}} + \frac{iwz_a}{\sqrt{K}} \right)}{2}$$

$$N_{1,-1}^{0,-} = \sqrt{2}\sqrt{P} \left(\frac{\sqrt{K}Pz_a\bar{z}_f}{2} - \frac{\sqrt{K}P\bar{z}_az_f}{2} - \sqrt{K}wz_a\bar{z}_f - \sqrt{K}w - \frac{iPRz_a^2\bar{z}_f}{2\sqrt{K}} + \frac{iPRz_f}{2\sqrt{K}} + \frac{Pwz_a^2\bar{z}_f^2}{2\sqrt{K}} - \frac{Pw}{2\sqrt{K}} - \frac{iRwz_a}{\sqrt{K}} - \frac{iRwz_f}{\sqrt{K}} + \frac{w^2z_a\bar{z}_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right)$$

$$N_{1,-1}^{1,+} = - \frac{\sqrt{2}\sqrt{P}R \left(i\sqrt{K}z_az_f + i\sqrt{K}z_a\bar{z}_f + \frac{Rz_f}{\sqrt{K}} + \frac{R\bar{z}_f}{\sqrt{K}} + \frac{iw\bar{z}_f^2}{\sqrt{K}} + \frac{iw}{\sqrt{K}} \right)}{2}$$

$$N_{1,-1}^{1,-} = \sqrt{2}\sqrt{P} \left(-\frac{\sqrt{K}Pz_a^2\bar{z}_f}{2} + \frac{\sqrt{K}Pz_f}{2} - \sqrt{K}wz_a - \sqrt{K}wz_f + \frac{iPRz_a\bar{z}_f}{2\sqrt{K}} - \frac{iPR\bar{z}_az_f}{2\sqrt{K}} - \frac{Pwz_a\bar{z}_f^2}{2\sqrt{K}} + \frac{Pw\bar{z}_a}{2\sqrt{K}} - \frac{iRwz_a\bar{z}_f}{\sqrt{K}} - \frac{iRw}{\sqrt{K}} + \frac{w^2z_a\bar{z}_f^2}{\sqrt{K}} + \frac{w^2\bar{z}_f}{\sqrt{K}} \right)$$

$$N_{1,-1}^{2,+} = \frac{\sqrt{2}\sqrt{P}R \left(-\sqrt{K}z_az_f - \sqrt{K}z_a\bar{z}_f - \frac{iRz_f}{\sqrt{K}} - \frac{iR\bar{z}_f}{\sqrt{K}} + \frac{w\bar{z}_f^2}{\sqrt{K}} + \frac{w}{\sqrt{K}} \right)}{2}$$

$$N_{1,-1}^{2,-} = \sqrt{2}\sqrt{P} \left(\frac{i\sqrt{K}Pz_a^2\bar{z}_f}{2} - \frac{i\sqrt{K}Pz_f}{2} + i\sqrt{K}wz_a + i\sqrt{K}wz_f - \frac{PRz_a\bar{z}_f}{2\sqrt{K}} + \frac{PR\bar{z}_az_f}{2\sqrt{K}} - \frac{iPwz_a\bar{z}_f^2}{2\sqrt{K}} + \frac{iPw\bar{z}_a}{2\sqrt{K}} + \frac{Rwz_a\bar{z}_f}{\sqrt{K}} + \frac{Rw}{\sqrt{K}} + \frac{iw^2z_a\bar{z}_f^2}{\sqrt{K}} + \frac{iw^2\bar{z}_f}{\sqrt{K}} \right)$$

$$N_{1,-1}^{3,+} = \frac{\sqrt{2}\sqrt{P}R \left(i\sqrt{K}z_f + i\sqrt{K}\bar{z}_f - \frac{Rz_az_f}{\sqrt{K}} - \frac{Rz_a\bar{z}_f}{\sqrt{K}} - \frac{iwz_a\bar{z}_f^2}{\sqrt{K}} - \frac{iwz_a}{\sqrt{K}} \right)}{2}$$

$$N_{1,-1}^{3,-} = \sqrt{2}\sqrt{P} \left(\frac{\sqrt{K}Pz_a\bar{z}_f}{2} - \frac{\sqrt{K}P\bar{z}_az_f}{2} - \sqrt{K}wz_a\bar{z}_f - \sqrt{K}w + \frac{iPRz_a^2\bar{z}_f}{2\sqrt{K}} - \frac{iPRz_f}{2\sqrt{K}} - \frac{Pwz_a^2\bar{z}_f^2}{2\sqrt{K}} + \frac{Pw}{2\sqrt{K}} + \frac{iRwz_a}{\sqrt{K}} + \frac{iRwz_f}{\sqrt{K}} - \frac{w^2z_a\bar{z}_f}{\sqrt{K}} - \frac{w^2}{\sqrt{K}} \right)$$

$$h = 1, \lambda = 1$$

$$N_{1,1}^{0,+} = \sqrt{2}\sqrt{P} \left(-\frac{i\sqrt{K}Rz_f}{2} - \frac{i\sqrt{K}R\bar{z}_f}{2} - \sqrt{K}wz_a\bar{z}_f + \sqrt{K}w - \frac{R^2z_az_f}{2\sqrt{K}} - \frac{R^2z_a\bar{z}_f}{2\sqrt{K}} - \frac{iRwz_a\bar{z}_f^2}{2\sqrt{K}} + \frac{iRwz_a}{2\sqrt{K}} - \frac{iRwz_f}{\sqrt{K}} - \frac{w^2z_a\bar{z}_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right)$$

$$\begin{aligned}
N_{1,1}^{0,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(-\sqrt{K}z_a\bar{z}_f + \sqrt{K}\bar{z}_az_f + \frac{iRz_a^2\bar{z}_f}{\sqrt{K}} - \frac{iRz_f}{\sqrt{K}} - \frac{wz_a^2\bar{z}_f^2}{\sqrt{K}} + \frac{w}{\sqrt{K}} \right)}{2} \\
N_{1,1}^{1,+} &= \sqrt{2}\sqrt{P} \left(\frac{i\sqrt{K}Rz_az_f}{2} + \frac{i\sqrt{K}Rz_a\bar{z}_f}{2} - \sqrt{K}wz_a + \sqrt{K}w\bar{z}_f + \frac{R^2z_f}{2\sqrt{K}} + \frac{R^2\bar{z}_f}{2\sqrt{K}} + \frac{iRwz_az_f}{\sqrt{K}} + \frac{iRw\bar{z}_f^2}{2\sqrt{K}} - \frac{iRw}{2\sqrt{K}} - \frac{w^2z_a}{\sqrt{K}} + \frac{w^2\bar{z}_f}{\sqrt{K}} \right) \\
N_{1,1}^{1,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(\sqrt{K}z_a^2\bar{z}_f - \sqrt{K}z_f - \frac{iRz_a\bar{z}_f}{\sqrt{K}} + \frac{iR\bar{z}_az_f}{\sqrt{K}} + \frac{wz_a\bar{z}_f^2}{\sqrt{K}} - \frac{w\bar{z}_a}{\sqrt{K}} \right)}{2} \\
N_{1,1}^{2,+} &= \sqrt{2}\sqrt{P} \left(\frac{\sqrt{K}Rz_az_f}{2} + \frac{\sqrt{K}Rz_a\bar{z}_f}{2} + i\sqrt{K}wz_a + i\sqrt{K}w\bar{z}_f + \frac{iR^2z_f}{2\sqrt{K}} + \frac{iR^2\bar{z}_f}{2\sqrt{K}} + \frac{Rwz_az_f}{\sqrt{K}} - \frac{Rw\bar{z}_f^2}{2\sqrt{K}} + \frac{Rw}{2\sqrt{K}} + \frac{iw^2z_a}{\sqrt{K}} + \frac{iw^2\bar{z}_f}{\sqrt{K}} \right) \\
N_{1,1}^{2,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(-i\sqrt{K}z_a^2\bar{z}_f + i\sqrt{K}z_f + \frac{Rz_a\bar{z}_f}{\sqrt{K}} - \frac{R\bar{z}_az_f}{\sqrt{K}} + \frac{iwz_a\bar{z}_f^2}{\sqrt{K}} - \frac{iw\bar{z}_a}{\sqrt{K}} \right)}{2} \\
N_{1,1}^{3,+} &= \sqrt{2}\sqrt{P} \left(-\frac{i\sqrt{K}Rz_f}{2} - \frac{i\sqrt{K}R\bar{z}_f}{2} + \sqrt{K}wz_a\bar{z}_f + \sqrt{K}w + \frac{R^2z_az_f}{2\sqrt{K}} + \frac{R^2z_a\bar{z}_f}{2\sqrt{K}} + \frac{iRwz_a\bar{z}_f^2}{2\sqrt{K}} - \frac{iRwz_a}{2\sqrt{K}} - \frac{iRwz_f}{\sqrt{K}} + \frac{w^2z_a\bar{z}_f}{\sqrt{K}} + \frac{w^2}{\sqrt{K}} \right) \\
N_{1,1}^{3,-} &= \frac{\sqrt{2}P^{\frac{3}{2}} \left(-\sqrt{K}z_a\bar{z}_f + \sqrt{K}\bar{z}_az_f - \frac{iRz_a^2\bar{z}_f}{\sqrt{K}} + \frac{iRz_f}{\sqrt{K}} + \frac{wz_a^2\bar{z}_f^2}{\sqrt{K}} - \frac{w}{\sqrt{K}} \right)}{2}
\end{aligned}$$

Proton-current components

$$s = -1, \quad s' = -1$$

$$\begin{aligned}
\mathcal{H}_0^{-1,-1} &= \frac{-\frac{iABEF_2\bar{z}_a}{m} - \frac{ABEF_2}{m} - \frac{iABE'F_2\bar{z}_a}{m} - \frac{ABE'F_2}{m} + 2iABF_1\bar{z}_a + 2ABF_1 + 2iABF_2\bar{z}_a + 2ABF_2 + \frac{iEF_2PR\bar{z}_a}{ABm} + \frac{EF_2PR}{ABm} + \frac{iE'F_2PR\bar{z}_a}{ABm} + \frac{E'F_2PR}{ABm} + \frac{2iF_1PR\bar{z}_a}{AB} + \frac{2F_1PR}{AB} + \frac{2iF_2PR\bar{z}_a}{AB} + \frac{2F_2PR}{AB}}{4} \\
\mathcal{H}_1^{-1,-1} &= \frac{\frac{ABF_2P\bar{z}_a}{m} + \frac{iABF_2P\bar{z}_a^2}{m} + \frac{ABF_2P\bar{z}_a}{m} + \frac{iABF_2P}{m} - \frac{4AF_1R\bar{z}_a}{B} - \frac{4iAF_1R}{B} - \frac{4AF_2R\bar{z}_a}{B} - \frac{4iAF_2R}{B} - \frac{4BF_1P\bar{z}_a}{A} - \frac{4iBF_1P}{A} - \frac{4BF_2P\bar{z}_a}{A} - \frac{4iBF_2P}{A} - \frac{F_2P^2R\bar{z}_a}{ABm} - \frac{iF_2P^2R\bar{z}_a^2}{ABm} - \frac{F_2P^2R\bar{z}_a}{ABm} - \frac{iF_2P^2R}{ABm}}{8}
\end{aligned}$$

$$\mathcal{H}_2^{-1,-1} = \frac{-\frac{iABF_2P\bar{z}_a}{m} - \frac{ABF_2P\bar{z}_a^2}{m} + \frac{iABF_2P\bar{z}_a}{m} + \frac{ABF_2P}{m} + \frac{2iABF_2R\bar{z}_a}{m} + \frac{2ABF_2R}{m} - \frac{4iAF_1R\bar{z}_a}{B} - \frac{4AF_1R}{B} - \frac{4iAF_2R\bar{z}_a}{B} - \frac{4AF_2R}{B} - \frac{4iBF_1P\bar{z}_a}{A} - \frac{4BF_1P}{A} - \frac{4iBF_2P\bar{z}_a}{A} - \frac{4BF_2P}{A} + \frac{iF_2P^2R\bar{z}_a}{ABm} + \frac{F_2P^2R\bar{z}_a^2}{ABm} - \frac{iF_2P^2R\bar{z}_a}{ABm} - \frac{F_2P^2R}{ABm} - \frac{2iF_2PR^2\bar{z}_a}{ABm} - \frac{2F_2PR^2}{ABm}}{8}$$

$$\mathcal{H}_3^{-1,-1} = \frac{\frac{iAF_1R\bar{z}_a}{B} - \frac{AF_1R}{B} + \frac{iAF_2R\bar{z}_a}{B} - \frac{AF_2R}{B} + \frac{iBF_1P\bar{z}_a}{A} - \frac{BF_1P}{A} + \frac{iBF_2P\bar{z}_a}{A} - \frac{BF_2P}{A}}{2}$$

$$s = -1, \quad s' = 1$$

$$\mathcal{H}_0^{1,-1} = \frac{\frac{ABEF_2\bar{z}_a}{m} + \frac{iABEF_2}{m} + \frac{ABE'F_2\bar{z}_a}{m} + \frac{iABE'F_2}{m} - 2ABF_1\bar{z}_a - 2iABF_1 - 2ABF_2\bar{z}_a - 2iABF_2 + \frac{EF_2PR\bar{z}_a}{ABm} + \frac{iEF_2PR}{ABm} + \frac{E'F_2PR\bar{z}_a}{ABm} + \frac{iE'F_2PR}{ABm} + \frac{2F_1PR\bar{z}_a}{AB} + \frac{2iF_1PR}{AB} + \frac{2F_2PR\bar{z}_a}{AB} + \frac{2iF_2PR}{AB}}{4}$$

$$\mathcal{H}_1^{1,-1} = \frac{-\frac{iABF_2Pz_a}{m} - \frac{ABF_2P\bar{z}_a^2}{m} - \frac{iABF_2P\bar{z}_a}{m} - \frac{ABF_2P}{m} - \frac{4iAF_1R\bar{z}_a}{B} - \frac{4AF_1R}{B} - \frac{4iAF_2R\bar{z}_a}{B} - \frac{4AF_2R}{B} + \frac{4iBF_1P\bar{z}_a}{A} + \frac{4BF_1P}{A} + \frac{4iBF_2P\bar{z}_a}{A} + \frac{4BF_2P}{A} - \frac{iF_2P^2Rz_a}{ABm} - \frac{F_2P^2R\bar{z}_a^2}{ABm} - \frac{iF_2P^2R\bar{z}_a}{ABm} - \frac{F_2P^2R}{ABm}}{8}$$

$$\mathcal{H}_2^{1,-1} = \frac{-\frac{ABF_2P\bar{z}_a}{m} - \frac{iABF_2P\bar{z}_a^2}{m} + \frac{ABF_2P\bar{z}_a}{m} + \frac{iABF_2P}{m} - \frac{2ABF_2R\bar{z}_a}{m} - \frac{2iABF_2R}{m} + \frac{4AF_1R\bar{z}_a}{B} + \frac{4iAF_1R}{B} + \frac{4AF_2R\bar{z}_a}{B} + \frac{4iAF_2R}{B} - \frac{4BF_1P\bar{z}_a}{A} - \frac{4iBF_1P}{A} - \frac{4BF_2P\bar{z}_a}{A} - \frac{4iBF_2P}{A} - \frac{F_2P^2Rz_a}{ABm} - \frac{iF_2P^2R\bar{z}_a^2}{ABm} + \frac{F_2P^2R\bar{z}_a}{ABm} + \frac{iF_2P^2R}{ABm} - \frac{2F_2PR^2\bar{z}_a}{ABm} - \frac{2iF_2PR^2}{ABm}}{8}$$

$$\mathcal{H}_3^{1,-1} = \frac{\frac{AF_1R\bar{z}_a}{B} - \frac{iAF_1R}{B} + \frac{AF_2R\bar{z}_a}{B} - \frac{iAF_2R}{B} - \frac{BF_1P\bar{z}_a}{A} + \frac{iBF_1P}{A} - \frac{BF_2P\bar{z}_a}{A} + \frac{iBF_2P}{A}}{2}$$

$$s = 1, \quad s' = -1$$

$$\mathcal{H}_0^{-1,1} = \frac{-\frac{ABEF_2z_a}{m} + \frac{iABEF_2}{m} - \frac{ABE'F_2z_a}{m} + \frac{iABE'F_2}{m} + 2ABF_1z_a - 2iABF_1 + 2ABF_2z_a - 2iABF_2 - \frac{EF_2PRz_a}{ABm} + \frac{iEF_2PR}{ABm} - \frac{E'F_2PRz_a}{ABm} + \frac{iE'F_2PR}{ABm} - \frac{2F_1PRz_a}{AB} + \frac{2iF_1PR}{AB} - \frac{2F_2PRz_a}{AB} + \frac{2iF_2PR}{AB}}{4}$$

$$\mathcal{H}_1^{-1,1} = \frac{\frac{ABF_2P\bar{z}_a^2}{m} - \frac{iABF_2Pz_a}{m} - \frac{iABF_2P\bar{z}_a}{m} + \frac{ABF_2P}{m} - \frac{4iAF_1Rz_a}{B} + \frac{4AF_1R}{B} - \frac{4iAF_2Rz_a}{B} + \frac{4AF_2R}{B} + \frac{4iBF_1Pz_a}{A} - \frac{4BF_1P}{A} + \frac{4iBF_2Pz_a}{A} - \frac{4BF_2P}{A} + \frac{F_2P^2Rz_a^2}{ABm} - \frac{iF_2P^2Rz_a}{ABm} - \frac{iF_2P^2R\bar{z}_a}{ABm} + \frac{F_2P^2R}{ABm}}{8}$$

$$\mathcal{H}_2^{-1,1} = \frac{-\frac{iABF_2P\bar{z}_a^2}{m} - \frac{ABF_2Pz_a}{m} + \frac{ABF_2P\bar{z}_a}{m} + \frac{iABF_2P}{m} + \frac{2ABF_2Rz_a}{m} - \frac{2iABF_2R}{m} - \frac{4AF_1Rz_a}{B} + \frac{4iAF_1R}{B} - \frac{4AF_2Rz_a}{B} + \frac{4iAF_2R}{B} + \frac{4BF_1Pz_a}{A} - \frac{4iBF_1P}{A} + \frac{4BF_2Pz_a}{A} - \frac{4iBF_2P}{A} - \frac{iF_2P^2Rz_a^2}{ABm} - \frac{F_2P^2Rz_a}{ABm} + \frac{F_2P^2R\bar{z}_a}{ABm} + \frac{iF_2P^2R}{ABm} + \frac{2F_2PR^2z_a}{ABm} - \frac{2iF_2PR^2}{ABm}}{8}$$

$$\mathcal{H}_3^{-1,1} = \frac{-\frac{AF_1Rz_a}{B} - \frac{iAF_1R}{B} - \frac{AF_2Rz_a}{B} - \frac{iAF_2R}{B} + \frac{BF_1Pz_a}{A} + \frac{iBF_1P}{A} + \frac{BF_2Pz_a}{A} + \frac{iBF_2P}{A}}{2}$$

$$s = 1, \quad s' = 1$$

$$\mathcal{H}_0^{1,1} = \frac{\frac{iABEF_2z_a}{m} - \frac{ABEF_2}{m} + \frac{iABE'F_2z_a}{m} - \frac{ABE'F_2}{m} - 2iABF_1z_a + 2ABF_1 - 2iABF_2z_a + 2ABF_2 - \frac{iEF_2PRz_a}{ABm} + \frac{EF_2PR}{ABm} - \frac{iE'F_2PRz_a}{ABm} + \frac{E'F_2PR}{ABm} - \frac{2iF_1PRz_a}{AB} + \frac{2F_1PR}{AB} - \frac{2iF_2PRz_a}{AB} + \frac{2F_2PR}{AB}}{4}$$

$$\mathcal{H}_1^{1,1} = \frac{-\frac{iABF_2P\bar{z}_a^2}{m} + \frac{ABF_2Pz_a}{m} + \frac{ABF_2P\bar{z}_a}{m} - \frac{iABF_2P}{m} - \frac{4AF_1Rz_a}{B} + \frac{4iAF_1R}{B} - \frac{4AF_2Rz_a}{B} + \frac{4iAF_2R}{B} - \frac{4BF_1Pz_a}{A} + \frac{4iBF_1P}{A} - \frac{4BF_2Pz_a}{A} + \frac{4iBF_2P}{A} + \frac{iF_2P^2Rz_a^2}{ABm} - \frac{F_2P^2Rz_a}{ABm} - \frac{F_2P^2R\bar{z}_a}{ABm} + \frac{iF_2P^2R}{ABm}}{8}$$

$$\mathcal{H}_2^{1,1} = \frac{-\frac{ABF_2P\bar{z}_a^2}{m} - \frac{iABF_2Pz_a}{m} + \frac{iABF_2P\bar{z}_a}{m} + \frac{ABF_2P}{m} - \frac{2iABF_2Rz_a}{m} + \frac{2ABF_2R}{m} + \frac{4AF_1Rz_a}{B} - \frac{4iAF_1R}{B} + \frac{4iAF_2Rz_a}{B} - \frac{4AF_2R}{B} + \frac{4iBF_1Pz_a}{A} - \frac{4iBF_1P}{A} + \frac{4iBF_2Pz_a}{A} - \frac{4BF_2P}{A} + \frac{F_2P^2Rz_a^2}{ABm} + \frac{iF_2P^2Rz_a}{ABm} - \frac{iF_2P^2R\bar{z}_a}{ABm} - \frac{F_2P^2R}{ABm} + \frac{2iF_2PR^2z_a}{ABm} - \frac{2F_2PR^2}{ABm}}{8}$$

$$\mathcal{H}_3^{1,1} = -\frac{\frac{iAF_1Rz_a}{B} + \frac{AF_1R}{B} + \frac{iAF_2Rz_a}{B} + \frac{AF_2R}{B} + \frac{iBF_1Pz_a}{A} + \frac{BF_1P}{A} + \frac{iBF_2Pz_a}{A} + \frac{BF_2P}{A}}{2}$$

All helicity amplitudes

Every nonzero amplitude is the following explicit four-component momentum-space scalar product:

$$\mathcal{M}_{h,s,s',\lambda} = \frac{1}{t} \sum_{\nu=0}^3 \left(\frac{N_{h\lambda}^{\nu,+}}{D_+} + \frac{N_{h\lambda}^{\nu,-}}{D_-} \right) \mathcal{H}_{\nu}^{s's}.$$

Spelling out the four terms for every surviving helicity channel gives

$$\begin{aligned} \mathcal{M}_{h=-1,s=-1,s'=-1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{-1,-1}^{0,+}}{D_+} + \frac{N_{-1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,-1} + \left(\frac{N_{-1,-1}^{1,+}}{D_+} + \frac{N_{-1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,-1} + \left(\frac{N_{-1,-1}^{2,+}}{D_+} + \frac{N_{-1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,-1} + \left(\frac{N_{-1,-1}^{3,+}}{D_+} + \frac{N_{-1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,-1} \right] \\ \mathcal{M}_{h=-1,s=-1,s'=-1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{-1,1}^{0,+}}{D_+} + \frac{N_{-1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,-1} + \left(\frac{N_{-1,1}^{1,+}}{D_+} + \frac{N_{-1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,-1} + \left(\frac{N_{-1,1}^{2,+}}{D_+} + \frac{N_{-1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,-1} + \left(\frac{N_{-1,1}^{3,+}}{D_+} + \frac{N_{-1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,-1} \right] \\ \mathcal{M}_{h=-1,s=-1,s'=1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{-1,-1}^{0,+}}{D_+} + \frac{N_{-1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,-1} + \left(\frac{N_{-1,-1}^{1,+}}{D_+} + \frac{N_{-1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,-1} + \left(\frac{N_{-1,-1}^{2,+}}{D_+} + \frac{N_{-1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,-1} + \left(\frac{N_{-1,-1}^{3,+}}{D_+} + \frac{N_{-1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,-1} \right] \\ \mathcal{M}_{h=-1,s=-1,s'=1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{-1,1}^{0,+}}{D_+} + \frac{N_{-1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,-1} + \left(\frac{N_{-1,1}^{1,+}}{D_+} + \frac{N_{-1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,-1} + \left(\frac{N_{-1,1}^{2,+}}{D_+} + \frac{N_{-1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,-1} + \left(\frac{N_{-1,1}^{3,+}}{D_+} + \frac{N_{-1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,-1} \right] \\ \mathcal{M}_{h=-1,s=1,s'=-1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{-1,-1}^{0,+}}{D_+} + \frac{N_{-1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,1} + \left(\frac{N_{-1,-1}^{1,+}}{D_+} + \frac{N_{-1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,1} + \left(\frac{N_{-1,-1}^{2,+}}{D_+} + \frac{N_{-1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,1} + \left(\frac{N_{-1,-1}^{3,+}}{D_+} + \frac{N_{-1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,1} \right] \\ \mathcal{M}_{h=-1,s=1,s'=-1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{-1,1}^{0,+}}{D_+} + \frac{N_{-1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,1} + \left(\frac{N_{-1,1}^{1,+}}{D_+} + \frac{N_{-1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,1} + \left(\frac{N_{-1,1}^{2,+}}{D_+} + \frac{N_{-1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,1} + \left(\frac{N_{-1,1}^{3,+}}{D_+} + \frac{N_{-1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,1} \right] \\ \mathcal{M}_{h=-1,s=1,s'=1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{-1,-1}^{0,+}}{D_+} + \frac{N_{-1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,1} + \left(\frac{N_{-1,-1}^{1,+}}{D_+} + \frac{N_{-1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,1} + \left(\frac{N_{-1,-1}^{2,+}}{D_+} + \frac{N_{-1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,1} + \left(\frac{N_{-1,-1}^{3,+}}{D_+} + \frac{N_{-1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,1} \right] \\ \mathcal{M}_{h=-1,s=1,s'=1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{-1,1}^{0,+}}{D_+} + \frac{N_{-1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,1} + \left(\frac{N_{-1,1}^{1,+}}{D_+} + \frac{N_{-1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,1} + \left(\frac{N_{-1,1}^{2,+}}{D_+} + \frac{N_{-1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,1} + \left(\frac{N_{-1,1}^{3,+}}{D_+} + \frac{N_{-1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,1} \right] \\ \mathcal{M}_{h=1,s=-1,s'=-1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{1,-1}^{0,+}}{D_+} + \frac{N_{1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,-1} + \left(\frac{N_{1,-1}^{1,+}}{D_+} + \frac{N_{1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,-1} + \left(\frac{N_{1,-1}^{2,+}}{D_+} + \frac{N_{1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,-1} + \left(\frac{N_{1,-1}^{3,+}}{D_+} + \frac{N_{1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,-1} \right] \\ \mathcal{M}_{h=1,s=-1,s'=-1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{1,1}^{0,+}}{D_+} + \frac{N_{1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,-1} + \left(\frac{N_{1,1}^{1,+}}{D_+} + \frac{N_{1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,-1} + \left(\frac{N_{1,1}^{2,+}}{D_+} + \frac{N_{1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,-1} + \left(\frac{N_{1,1}^{3,+}}{D_+} + \frac{N_{1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,-1} \right] \\ \mathcal{M}_{h=1,s=-1,s'=1,\lambda=-1} &= \frac{1}{t} \left[\left(\frac{N_{1,-1}^{0,+}}{D_+} + \frac{N_{1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,-1} + \left(\frac{N_{1,-1}^{1,+}}{D_+} + \frac{N_{1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,-1} + \left(\frac{N_{1,-1}^{2,+}}{D_+} + \frac{N_{1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,-1} + \left(\frac{N_{1,-1}^{3,+}}{D_+} + \frac{N_{1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,-1} \right] \\ \mathcal{M}_{h=1,s=-1,s'=1,\lambda=1} &= \frac{1}{t} \left[\left(\frac{N_{1,1}^{0,+}}{D_+} + \frac{N_{1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,-1} + \left(\frac{N_{1,1}^{1,+}}{D_+} + \frac{N_{1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,-1} + \left(\frac{N_{1,1}^{2,+}}{D_+} + \frac{N_{1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,-1} + \left(\frac{N_{1,1}^{3,+}}{D_+} + \frac{N_{1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,-1} \right] \end{aligned}$$

$$\mathcal{M}_{h=1,s=1,s'=-1,\lambda=-1} = \frac{1}{t} \left[\left(\frac{N_{1,-1}^{0,+}}{D_+} + \frac{N_{1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,1} + \left(\frac{N_{1,-1}^{1,+}}{D_+} + \frac{N_{1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,1} + \left(\frac{N_{1,-1}^{2,+}}{D_+} + \frac{N_{1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,1} + \left(\frac{N_{1,-1}^{3,+}}{D_+} + \frac{N_{1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,1} \right]$$

$$\mathcal{M}_{h=1,s=1,s'=-1,\lambda=1} = \frac{1}{t} \left[\left(\frac{N_{1,1}^{0,+}}{D_+} + \frac{N_{1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{-1,1} + \left(\frac{N_{1,1}^{1,+}}{D_+} + \frac{N_{1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{-1,1} + \left(\frac{N_{1,1}^{2,+}}{D_+} + \frac{N_{1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{-1,1} + \left(\frac{N_{1,1}^{3,+}}{D_+} + \frac{N_{1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{-1,1} \right]$$

$$\mathcal{M}_{h=1,s=1,s'=1,\lambda=-1} = \frac{1}{t} \left[\left(\frac{N_{1,-1}^{0,+}}{D_+} + \frac{N_{1,-1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,1} + \left(\frac{N_{1,-1}^{1,+}}{D_+} + \frac{N_{1,-1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,1} + \left(\frac{N_{1,-1}^{2,+}}{D_+} + \frac{N_{1,-1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,1} + \left(\frac{N_{1,-1}^{3,+}}{D_+} + \frac{N_{1,-1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,1} \right]$$

$$\mathcal{M}_{h=1,s=1,s'=1,\lambda=1} = \frac{1}{t} \left[\left(\frac{N_{1,1}^{0,+}}{D_+} + \frac{N_{1,1}^{0,-}}{D_-} \right) \mathcal{H}_0^{1,1} + \left(\frac{N_{1,1}^{1,+}}{D_+} + \frac{N_{1,1}^{1,-}}{D_-} \right) \mathcal{H}_1^{1,1} + \left(\frac{N_{1,1}^{2,+}}{D_+} + \frac{N_{1,1}^{2,-}}{D_-} \right) \mathcal{H}_2^{1,1} + \left(\frac{N_{1,1}^{3,+}}{D_+} + \frac{N_{1,1}^{3,-}}{D_-} \right) \mathcal{H}_3^{1,1} \right]$$

The remaining 16 amplitudes obey $\mathcal{M}_{h' \neq h, s, s', \lambda} = 0$ exactly.